Retention playbook

How to run one retention wave with the recommended rule: who to contact, with which offer, through which channel, at what budget, and how to know whether it worked.

1. Objective and rule

Objective: maximise expected net retention value from a fixed budget while keeping contact fair, lawful and measurable.

Rule: Expected value + segment offers. For every customer and every eligible offer:

> Expected net value (EV) = p(churn) x save rate x customer value - (offer cost + contact cost)
>
> Customer value = monthly charges x margin 30% x discounted months (base 12 months x tenure factor 0.75/1.0/1.25, 10% a year)

The optimiser picks at most one offer per customer to maximise total EV within the budget, never contacts a customer with EV <= 0, and never contacts more than 30% of customers in a wave.

Worked example: a month-to-month customer paying $70 a month with 20 months of tenure (factor 1.0) is worth $70 x 30% x 11.40 discounted months = **$239**. With a churn probability of 60% and the standard offer (save rate 20%, cost $25), EV = 0.6 x 0.2 x $239 - $25 = **$3.73**. At this risk level the offer pays only for customers worth more than $208.

2. Who to contact (campaign size)

Plan	Customers contacted (test sample of 1,409)	Budget	Expected net value	ROI
Base budget (standard offer for 10% of customers)	153 (11%)	$3,500	$2,668	76%
Recommended (standard offer for 26% of customers, validation rule)	**324 (23%)**	**$9,150**	**$4,145**	**45%**
All positive-EV customers	371 (26%)	$11,318	see profit curve	lower marginal return

Scaled to the full base of 7,043 customers (linear, indicative): a full wave is about 1,620 offers, $45,737 budget and $20,721 expected net value. The pilot uses a list of the same size but makes offers to only half of it (810 customers); the other half is the control group.

Budget tranches. Release budget in steps and stop when the marginal ROI of the next step falls below 10%. The budget share is the share of customers the standard offer ($25) could reach; the optimiser mixes offers, so the number contacted differs:

Budget share	Customers contacted	Cumulative cost	Cumulative expected net value	Marginal ROI of the step
4%	71	$1,400	$1,417	83%
8%	124	$2,800	$2,308	59%
12%	178	$4,225	$2,976	42%
16%	226	$5,625	$3,457	32%
20%	269	$7,025	$3,810	23%
24%	300	$8,450	$4,057	15%
28%	340	$9,849	$4,210	9%
32%	369	$11,246	$4,256	2%

3. What to offer

Offer	Cost + contact	Assumed save rate	Eligible	Recommended wave (test)	Pilot wave, offered half (about)	Base-budget wave (test)

One-time $20 bill credit	$25	20%	All customers	24	60	6
$35 credit for moving to a 1-year contract	$40	30%	Contract: Month-to-month	150	375	37
6 months free TechSupport + OnlineSecurity	$17	15%	InternetService: DSL or Fiber optic; TechSupport: No	150	375	110
$10 loyalty reward	$15	10%	tenure_segment: loyal	0	0	0

The pilot column scales the recommended wave to the pilot's offered half; each offer is its own sub-arm, and the first decision is about the list rule as a whole (pilot design).

4. How and when to contact

The rules below are those a Vietnamese operator would apply; the deployment note explains the legal and price context. A US operator would apply the equivalent national rules.

1. **Consent and suppression first.** Contact only customers with recorded, purpose-specific consent for marketing; scrub every list against the national Do-Not-Call register and internal opt-outs before each wave (Decree 91/2020, PDP Law 2025).
2. **Channel:** one outbound call by a trained retention agent, followed by an SMS confirming the offer with a clear opt-out. Calls only **08:00-17:00**, SMS only 08:00-20:00 (stricter than the 07:00-22:00 legal window because aggregators block evening promotions).
3. **Frequency:** at most **one advertising call per customer per 24 hours** by law; internal rule: one retention contact per customer per 30 days, and no new offer within 90 days of a previous one.
4. **Script:** acknowledge the relationship, ask what would make the customer stay, present one offer (the assigned one), no pressure, record acceptance, refusal or "do not contact".
5. **Log the reason and route service problems.** In the sample only 46% of churners left for price or a competitor's offer; 42% left over staff attitude or dissatisfaction with the network, product or support (analysis report). Agents record the customer's reason with a fixed code list and pass service complaints to the responsible team instead of answering them with a discount.
6. **Brand name** on every SMS; stop immediately on refusal; keep consent and refusal logs for at least one year.

5. Measurement

Every wave keeps a **random holdout** inside the target list (pilot design):

- 50% of eligible customers in the pilot wave are not contacted;
- primary outcome: churn within 90 days, contacted vs holdout;
- measured save rate = 1 - churn(offered) / churn(holdout), for the list and per offer;
- update configs/default.yaml with the measured save rates and re-run the plan.

6. Stop / go rules

Signal	Action
Measured save rate: 95% interval above 0 and estimate at least 15.6% (break-even of the recommended list)	Go: scale to the recommended budget
Estimate at or below 0 (contacted churn >= holdout churn), or whole interval below break-even	Stop; redesign offers and targeting (consider uplift modelling)
Anything else	Extend the pilot (another wave, pooled), at most 2 times; then stop
Stop decided	Keep the base budget on the expected-value list with the cheapest offer that the pilot did not rule out, or return it to Finance; fix the service reasons customers give for leaving
Opt-out or complaint rate > 2x the normal campaign rate	Pause and review script and frequency
Fairness review flag on the wave list	DPO and Retention owner review before launch

7. Customer-experience and fairness guardrails

- Protected attributes (gender, age) are never used to choose customers or offers; their selection rates are audited every wave.
- Senior customers are contacted 3.3x as often as others (2.6x under risk-only targeting) and 1.7x after adjusting for churn risk, so higher risk explains only part of the gap; the rest comes from higher bills. Agents follow accessible-communication guidance and use no high-pressure tactics, and the pilot reports seniors separately.
- Low-risk loyal customers rarely qualify for retention offers, and the loyalty reward in the catalog is not chosen by the optimiser under current assumptions. To avoid a perceived "loyalty penalty", pair the campaign with a non-targeted loyalty benefit funded outside the retention budget.
- Customers who decline are suppressed from retention offers for 90 days.

8. Operating checklist per wave

1. Score the current customer base with the deployed model (dashboard: *Model evidence -> Data and new customers*, or python -m telco_retention score).
2. Confirm assumptions with Finance (margin, offer cost) and Operations (contact cost, capacity) on the *Assumptions & sensitivity* page; check the chance of a loss.
3. Review the list on the *Decision* page and export the campaign ZIP (list + assumptions + scenario ID).
4. Apply consent / DNC suppression; randomise the holdout; freeze and archive the list.
5. Execute contacts within the legal windows; log outcomes daily.
6. After 90 days, measure churn by arm and offer, update the assumption register, decide go / extend / stop.